

# Electoral Accountability and Economic Policy

Based on Besley & Case (1995)

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- A central question in political economy: **do electoral incentives discipline politicians?**
- Do politicians act in the public interest, or do they use political power for personal gain?
- The desire to maintain a reputation is often thought to keep politicians in check
- Officials who care to run again must act in voters' interest to merit reelection
- But what happens when reelection is no longer possible?
- **Besley & Case (1995)** exploit gubernatorial term limits in U.S. states to test this

# Theoretical Framework: Reputation-Building Model

- Based on Banks & Sundaram (1993)
- Each governor has an unobservable **type**  $\omega \in \{\omega_1, \dots, \omega_N\}$
- Takes unobservable **effort**  $\alpha$ , which affects policy outcome  $r$
- Voters observe  $r$  and use it to infer the governor's type via Bayes' rule

## Two Regimes

- 1 **Term-limited (lame duck)**: maximizes immediate payoffs — puts in less effort
- 2 **Reelection-eligible**: puts in extra effort to build reputation and secure reelection

# Relevance for Our Paper

Besley & Case (1995) establish:

- ① Reelection incentives discipline politicians' fiscal behavior
- ② Term limits provide exogenous variation in electoral incentives
- ③ The effect operates through reputation-building, not party discipline

## Our contribution

We apply this framework to **Brazilian municipalities** in São Paulo, asking whether reelection eligibility affects **spending composition** — not just the level of taxes and spending — and whether this effect is moderated by **fiscal discretion**.

# Brazilian Municipal Elections

- Brazilian municipalities hold **direct elections** for mayor every four years
- Elections are held simultaneously across all municipalities — in our sample, the relevant elections occurred in **2016** and **2020**
- São Paulo state alone has **645 municipalities**, ranging from the capital with over 12 million inhabitants to small rural towns with fewer than 1,000 residents

# Term Limits in Brazilian Municipalities

- Brazilian law limits mayors to **two consecutive terms** in office
- A mayor elected in 2016 for the **first time** is eligible to run again in 2020
- A mayor elected in 2016 who was **already serving their first term** since 2012 faces a binding term limit and cannot run in 2020
- This generates clean variation in reelection incentives **within the same election cycle** across municipalities

## Identification

Comparing mayors who can run again in 2020 against those who cannot, conditional on controls, isolates the causal effect of reelection incentives on spending behavior.

# Who Controls Municipal Spending?

- Brazilian municipalities operate under a **separation of powers**: the *Câmara dos Vereadores* (city council) approves laws and the annual budget, while the **mayor executes it**
- In practice, the mayor has **substantial discretion** over spending composition through several channels:
  - ① **Budget proposal**: the mayor drafts the annual budget (*Lei Orçamentária Anual*), which the council typically approves with minor modifications
  - ② **Execution discretion**: the mayor controls the *timing* and *allocation* of expenditures within approved categories
  - ③ **Supplementary credits**: the mayor can open *créditos suplementares* to reallocate spending across categories without full council approval
  - ④ **Administrative decisions**: hiring, procurement, and contracting decisions — which shape spending composition in practice — are executive prerogatives

# The Limits of Mayoral Discretion

- Not all spending is equally subject to mayoral discretion
- A large share of municipal revenue is **constitutionally earmarked**:
  - **FUNDEB**: at least 25% of tax revenue must be spent on education
  - **SUS transfers**: federal health transfers are earmarked exclusively to health spending
  - **Convênios**: federal and state grants are tied to specific programs
- What remains after earmarked transfers is **treasury revenue** (*tesouro*) — the discretionary budget the mayor truly controls
- This motivates our key moderating variable: the **share of total spending funded by discretionary treasury resources**,  $D_{it}$ , which measures how much fiscal power the mayor actually has

# Our Research Question

## Central Question

Do reelection incentives affect how Brazilian mayors allocate their **discretionary** budget across spending categories?

- We follow Besley & Case (1995) in using **term limits** as a source of exogenous variation in reelection incentives
- Our contribution is to focus on **spending composition** rather than aggregate levels, and to explicitly account for **fiscal discretion** as a moderating force
- The Brazilian context is particularly informative because **heavy earmarking** limits mayoral discretion — if political budget cycles exist, they should be concentrated in the discretionary budget

- **Municipal expenditure data:** Tribunal de Contas do Estado de São Paulo (TCESP)
  - Annual expenditure records by spending category and funding source
  - 643 municipalities, 2017–2024
- **Electoral data:** Tribunal Superior Eleitoral (TSE)
  - Candidate-level data for 2016 and 2020 municipal elections
  - Reelection eligibility, vote shares, number of candidates
- **Socioeconomic controls:** IBGE Census 2010 and Atlas Brasil
  - Population, PIB per capita, age distribution
  - Municipal HDI, sanitation access, infant mortality
- All socioeconomic controls are measured in 2010, eliminating reverse causality concerns.

# Dependent Variable: Discretionary Spending Share

- Standard measures of municipal spending conflate **mayoral choices** with **mechanically earmarked transfers**
- We focus instead on the **discretionary spending share**:

## Dependent Variable

$$s_{ikt}^d = \frac{\text{Discretionary spending in category } k_{it}}{\text{Total discretionary spending}_{it}} \times 100$$

- **Discretionary spending** = expenditure funded by *tesouro* or *recursos próprios da administração indireta*
- **Earmarked spending** = federal/state transfers and special funds (*FUNDEB, SUS, convênios*)
- $s_{ikt}^d$  measures **how much of total discretionary money did the mayor choose to put in a given category** — a genuine revealed preference for spending priority.

# Key Variables

- **Treatment variable**  $R_{it} \in \{0, 1\}$ :
- $R_{it} \in \{0, 1\}$  varies at the **municipality-term** level:
  - For 2017–2020:  $R_{i_1} = 1$  if the 2016 winner was in their **first term**, eligible to run in 2020;  $R_{i_1} = 0$  if they had already served since 2012
  - For 2021–2024:  $R_{i_2} = 1$  if the 2020 winner was in their **first term**, eligible to run in 2024;  $R_{i_2} = 0$  if they had already served since 2016
- Predetermined relative to spending: determined by prior election outcomes before each term begins

- **Moderating variable**  $D_{it}$ :

- Discretionary share of total expenditure:

$$D_{it} = \frac{\text{Total discretionary spending}_{it}}{\text{Total spending}_{it}} \times 100$$

- Measures **mayoral fiscal power** — how much of the total budget the mayor truly controls
- Varies at the municipality-year level as federal transfer rules and revenues change

# Regression Equation

## Estimated Model

$$s_{ikt}^d = \beta_0 + \beta_1 R_{it} + \beta_2 \tilde{D}_{it} + \beta_3 (R_{it} \times \tilde{D}_{it}) + \beta_4 E_{it} + \beta_5 N_{it} + \mathbf{X}_i' \gamma + \mu_t + \varepsilon_{it}$$

- $s_{ikt}^d$ : discretionary spending share in category  $k$ , municipality  $i$ , year  $t$
- $R_{it}$ : reelection eligibility — varies at municipality-term level
- $\tilde{D}_{it}$ : demeaned discretionary share of total expenditure
- $R_{it} \times \tilde{D}_{it}$ : interaction — does greater fiscal discretion amplify the effect of reelection incentives on spending composition?
- $E_{it}$ : total expenditure per capita
- $N_{it}$ : number of candidates in the relevant election — varies at municipality-term level
- $\mathbf{X}_i$ : 2010 Census controls (PIB per capita, IDHM, share of children)
- $\mu_t$ : year fixed effects

# Coefficients of Interest

- $\beta_1$  — **main effect of reelection eligibility:**
  - Effect of reelection eligibility on discretionary spending composition **at the sample mean of fiscal discretion** ( $\bar{D} = 71.4\%$ )
  - If  $\beta_1 \neq 0$ : reelection incentives distort discretionary spending composition
  - If  $\beta_1 = 0$ : no political budget cycle in discretionary spending
- $\beta_2$  — **effect of fiscal discretion on spending composition:**
  - How spending shares change as fiscal discretion moves **above or below the sample mean** of 71.4%
  - Positive  $\beta_2$ : above average fiscal discretion  $\rightarrow$  more spending in that category
  - Negative  $\beta_2$ : above average fiscal discretion  $\rightarrow$  less spending in that category

- $\beta_3$  — **moderation effect of fiscal discretion on reelection incentives:**
  - How the spending gap between reelection-eligible and term-limited mayors changes as fiscal discretion moves above or below its sample mean
  - Negative  $\beta_3$ : gap **narrows** for above-average discretion municipalities
  - Positive  $\beta_3$ : gap **widens** for above-average discretion municipalities

# Spending Categories

- We estimate the model separately for **six functional spending categories** from TCESP data

| Category                      | Description                                               |
|-------------------------------|-----------------------------------------------------------|
| Health ( <i>Saúde</i> )       | Primary healthcare, hospitals, and public health programs |
| Education ( <i>Educação</i> ) | Schools, teachers, and educational infrastructure         |
| Urbanismo                     | Urban infrastructure, roads, and public works             |
| Social Assistance             | Welfare programs and social protection                    |
| Administration                | General government overhead and bureaucracy               |
| Transport                     | Public transportation and road maintenance                |

- Health and education are heavily constrained by **constitutional earmarking** (SUS, FUNDEB) — the reelection effect and its moderation by fiscal discretion are expected to be concentrated in the remaining categories where mayors have genuine room to maneuver

- We estimate the model separately for six spending categories
- Results are organized around three findings:
  - ① **Reelection eligibility** ( $\beta_1$ ) significantly affects discretionary spending in **urbanismo** and **administration**
  - ② **Fiscal discretion** ( $\beta_2$ ) shapes spending composition independently of electoral incentives in **social assistance**, **transport**, and **urbanismo**
  - ③ The **moderation effect** ( $\beta_3$ ) is statistically significant in **social assistance** and **administration**, but economically negligible
- Health, education, and transport show **no significant reelection effect** — consistent with constitutional earmarking limiting mayoral discretion in these categories

# Finding 1: Reelection Eligibility and Spending Composition

| Category       | Coefficient on $R_{it}$ ( $\beta_1$ ) |            |        |         |
|----------------|---------------------------------------|------------|--------|---------|
|                | Estimate                              | Std. Error | T-stat | P-value |
| Health         | -0.093                                | 0.279      | -0.34  | 0.738   |
| Education      | 0.065                                 | 0.199      | 0.32   | 0.746   |
| Urbanismo      | -0.963***                             | 0.299      | -3.22  | 0.001   |
| Social Assist. | 0.080                                 | 0.100      | 0.80   | 0.425   |
| Administration | 1.027***                              | 0.373      | 2.75   | 0.006   |
| Transport      | 0.023                                 | 0.201      | 0.11   | 0.910   |

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ . Clustered SE at municipality level.

- Reelection eligibility affects spending composition **only through urbanismo and administration**

# Finding 1: Reelection Eligibility and Urbanismo

## Result

Reelection-eligible mayors allocate **0.96 percentage points less** of their discretionary budget to urbanismo relative to term-limited mayors ( $\beta_1 = -0.963$ ,  $p = 0.001$ )

- Urbanismo covers **urban infrastructure, roads, and public works** — a category historically associated with corruption in Brazilian municipalities
- We interpret the **negative  $\beta_1$**  as evidence of **electoral accountability constraining opportunistic spending**: reelection-eligible mayors restrain infrastructure spending because corruption in public works carries electoral risk before the next election

# Finding 1: Reelection Eligibility and Urbanismo

## Result

Reelection-eligible mayors allocate **0.96 percentage points less** of their discretionary budget to urbanismo relative to term-limited mayors ( $\beta_1 = -0.963$ ,  $p = 0.001$ )

- Term-limited mayors, facing **no future electoral punishment**, increase discretionary urbanismo spending — consistent with rent extraction through inflated construction contracts, a well-documented phenomenon in Brazilian public administration
- This is consistent with the **reputation-building model**: accountability disciplines mayors when reelection is at stake, and opportunistic behavior emerges when it is not

# Finding 1: Reelection Eligibility and Administration

## Result

Reelection-eligible mayors allocate **1.03 percentage points more** of their discretionary budget to administration relative to term-limited mayors ( $\beta_1 = 1.027$ ,  $p = 0.006$ )

- Administration covers **general government overhead**, including public hiring and bureaucratic expenditure
- We interpret this as evidence of **political patronage**: reelection-seeking mayors strategically expand the municipal payroll to generate electoral loyalty

# Finding 1: Reelection Eligibility and Administration

## Result

Reelection-eligible mayors allocate **1.03 percentage points more** of their discretionary budget to administration relative to term-limited mayors ( $\beta_1 = 1.027$ ,  $p = 0.006$ )

- This is consistent with the broader literature on **political clientelism** in Brazilian municipalities, where public employment is a well-documented tool of electoral mobilization
- Importantly, this effect is identified **within the discretionary budget** — it reflects a genuine mayoral choice to prioritize administrative hiring over service delivery when reelection is at stake

## Finding 2: Fiscal Discretion and Spending Composition

### Result

The discretionary share of total expenditure ( $\tilde{D}_{it}$ ) significantly shapes spending composition in three categories, independently of electoral incentives

| Category       | Estimate  | Std. Error | T-stat | P-value |
|----------------|-----------|------------|--------|---------|
| Social Assist. | +0.054*** | 0.015      | 3.63   | 0.000   |
| Transport      | +0.069*** | 0.025      | 2.74   | 0.006   |
| Urbanismo      | −0.095**  | 0.041      | −2.33  | 0.020   |

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

## Finding 2: Fiscal Discretion and Spending Composition

### Result

The discretionary share of total expenditure ( $\tilde{D}_{it}$ ) significantly shapes spending composition in three categories, independently of electoral incentives

- Municipalities with **above average fiscal discretion** allocate proportionally more to social assistance and transport — when mayors have more budget freedom than the typical municipality, these categories reflect **genuine mayoral spending preferences**
- Municipalities with **above average fiscal discretion** allocate proportionally **less** to urbanismo — when mayors have more fiscal freedom, infrastructure competes less successfully for discretionary funds as other categories absorb the additional room to maneuver

## Finding 2: Fiscal Discretion and Spending Composition

### Result

The discretionary share of total expenditure ( $\tilde{D}_{it}$ ) significantly shapes spending composition in three categories, independently of electoral incentives

- **Robustness check:** Fiscal discretion is **not simply a proxy for municipal wealth** — the correlation between  $\tilde{D}_{it}$  and PIB per capita is only 0.12, and with IDHM only 0.21

## Finding 3: The Moderation Effect

### Result

The interaction  $R_{it} \times \tilde{D}_{it}$  reveals **heterogeneity in the form of electoral manipulation** depending on the level of fiscal discretion

| Category       | Estimate  | Std. Error | T-stat | P-value |
|----------------|-----------|------------|--------|---------|
| Health         | +0.079*   | 0.047      | 1.68   | 0.094   |
| Social Assist. | -0.040*** | 0.014      | -2.84  | 0.005   |
| Administration | -0.149**  | 0.057      | -2.60  | 0.009   |

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

## Finding 3: Conditional Clientelism

### Unified Interpretation

The form of electoral manipulation depends on **how much fiscal room a mayor has**

- **Mayors with below average fiscal discretion** rely on **two electoral instruments simultaneously**:
  - **Patronage hiring** — expanding the municipal payroll generates electoral loyalty when discretionary funds are scarce, with the effect growing stronger as discretion falls below the mean
  - **Social assistance** — targeting direct transfers to voters becomes the preferred electoral tool when fiscal room is tight
- **Mayors with above average fiscal discretion** show **attenuated effects** in both categories — at one standard deviation above the mean, the administration effect (+1.027 pp) is nearly fully offset (−1.12 pp), suggesting that patronage hiring becomes unnecessary when mayors have ample budget room

## Finding 3: Conditional Clientelism

### Unified Interpretation

The form of electoral manipulation depends on **how much fiscal room a mayor has**

- This is consistent with a **conditional clientelism** story: fiscal constraint shapes the instruments of electoral manipulation, with cash-strapped mayors relying more heavily on patronage and direct transfers

# The Null Results: Health and Education

- Reelection eligibility has **no significant effect** on discretionary spending shares for health or education
- This is consistent with **constitutional earmarking** constraining mayoral choices even within the discretionary budget:
  - FUNDEB mandates minimum education spending — mayors face political and legal costs from deviating
  - SUS transfers create institutional pressures to maintain health spending levels
- The null results are a **substantive finding**, not a failure of the model — they confirm that institutional constraints on spending are binding even at the level of discretionary allocation
- This contributes to the literature on **fiscal federalism and political accountability**: earmarking effectively insulates key social spending from electoral manipulation, even when mayors have discretionary resources available